

Technical Case - Frontend Engineer

A company operates a fleet of meeting-room devices across its offices and currently has no single place to view them. Your task is to build that dashboard: a view of the status of every device, with the ability to manage the fleet from one screen. This is a frontend task, to be built in React and TypeScript, and scoped for four to six hours.

Requirements

Device List

- List each device and show name, model and status (*online, in meeting, offline, deactivated*).
- Filter the list by name. (*optional*)

Device Management

- Add a device via a form with Name, Description and Status. Saving should update both the list and the summary counts.
- Remove a device from the list.

Chart (*optional, but a plus*)

- Device status over time. Any library, any granularity.

Data and API Guidelines

Two JSON files are attached so that you do not have to spend time on creating mock data:

- `devices.json` : 433 devices, with the same status counts shown in the reference design.
- `Status-history.json` : monthly status counts, for the chart.

These are provided for convenience and are not a schema. The data model is your decision: reshape, rename or discard whatever you like.

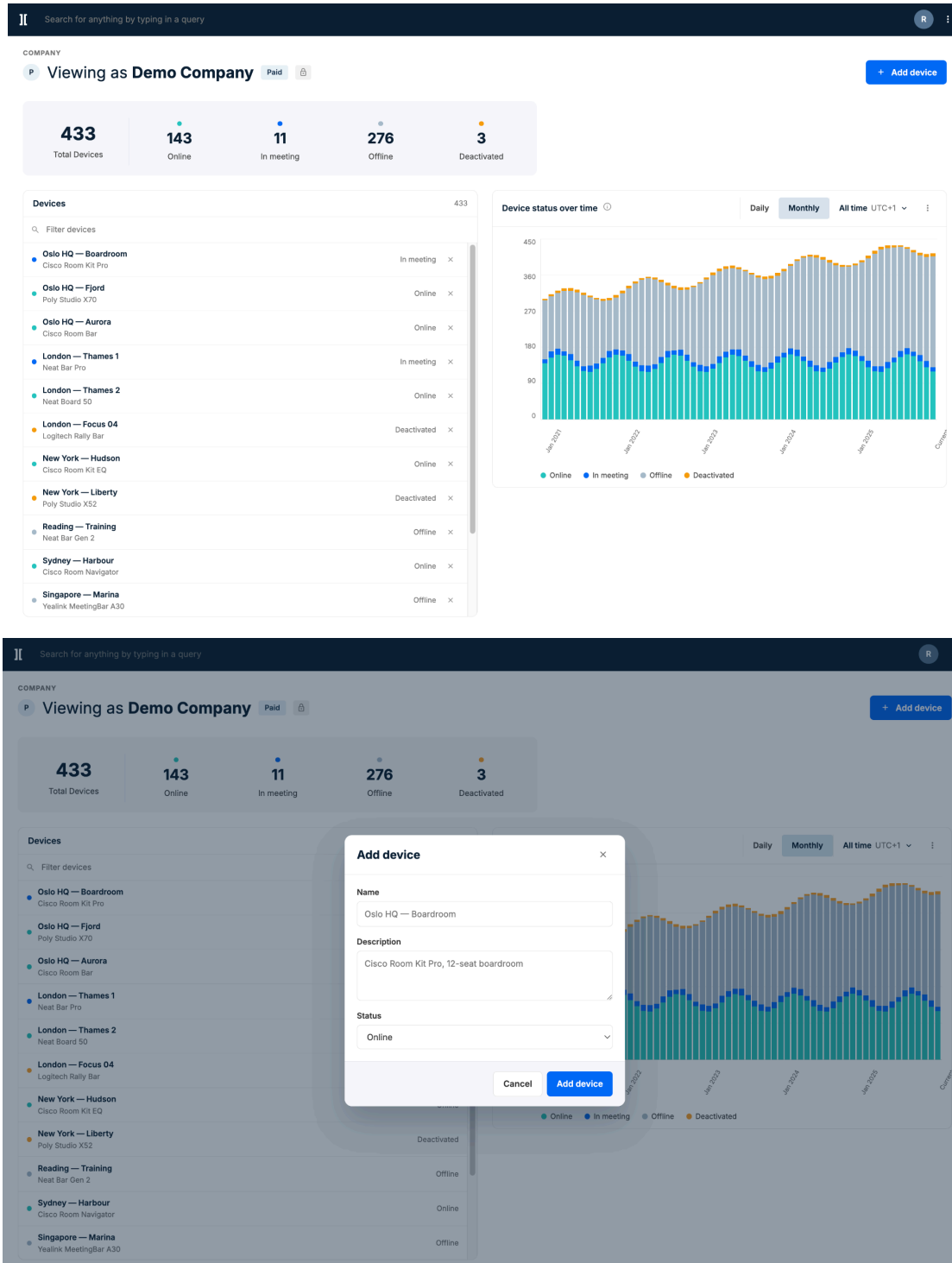
Example API (*Illustrative only*).

```
GET  /api/devices      → [{ id, name, model, description, status }]
POST /api/devices      { name, description, status }
PATCH /api/devices/:id { name?, description?, status? }
DELETE /api/devices/:id
GET  /api/devices/summary → { total, online, inMeeting, offline, deactivated }
GET  /api/devices/history → [{ period, online, inMeeting, offline, deactivated }]
```

Deriving the summary on the client instead is a perfectly good call.

Visual Inspiration

The attached screenshots show one possible layout. They are purely for inspiration, you are not expected to mirror the visual design or styling, and we do not assess pixel accuracy.



Notes

You will be presenting your work towards this project during the technical interview process. How you reason about the choices and decisions you've made, matters more than anything else in this brief.

You are welcome to use AI tools, but the work you submit is yours to take ownership of. The follow-up conversation will focus on the decisions in your code, why you structured it as you did, what tradeoffs you made, and what you would change with more time.

What we're looking for

- **A conversation starter, not a finished product:** We want a working attempt to anchor our technical discussion, a partial or half-finished solution is completely fine provided you stick to the 4–6 hour time box. Do not over-invest time polishing this.
- **Reasoning over completeness:** How you think through problems, justify your architectural choices, and structure your code matters far more than shipping every requirement.
- **Clean fundamentals:** Clear, readable React and TypeScript with sensible component boundaries.
- **Trade-off awareness:** A clear understanding of what you built, why you made specific choices, and what you would prioritize next given more time.

Any questions please feel free to ask me over email.
Good luck, and thanks for taking the time.